

Zero-Shot Polyp Detection in Open-World Endoscopy via A Cascaded Detector-Verifier Framework

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Abstract

Polyp detectors trained on curated endoscopic datasets often underperform in real clinical environments, where images may suffer from degradations that are difficult to collect and annotate. Consequently, models are frequently deployed in zero-shot scenarios with unseen degradations, a setting that has received limited attention. In this work, we address this gap by creating a synthetic degraded benchmark derived from CVC-ClinicDB and Kvasir-SEG using Qwen-Image-Edit, enabling systematic zero-shot evaluation. In addition, we propose a detector-verifier framework that combines an object detector with a vision-language model (VLM) verifier, which leverages semantic reasoning to adaptively confirm candidate detections. The verifier is further fine-tuned using Group Relative Policy Optimization (GRPO) to penalize missed detections. Experiments on degraded image sets show substantial recall improvement over the detector alone while maintaining comparable precision, whereas performance on clean images remains similar to the baseline. These results demonstrate our approach provides improved clinical alignment with fewer false negatives in open-world polyp detection, reducing the risk of missed precancerous polyps for improved patient outcomes.

1. Introduction

The efficacy of early screening for colorectal cancer depends critically on accurate polyp detection [5]. Public endoscopic datasets, such as CVC-ClinicDB [4] and Kvasir-SEG [12], primarily contain clean, well-lit images and fail to capture the diversity of imaging conditions encountered in real-world clinical practice. Models trained on these ide-

alized datasets tend to overfit, resulting in poor generalization to unseen clinical centers, imaging modalities, or degraded acquisition scenarios [2, 18]. Collecting annotated data that covers all possible variations is practically infeasible, making it challenging to ensure reliable detection under highly variable gastrointestinal imaging conditions [29]. Consequently, models are often deployed in a zero-shot scenario, where they must detect polyps in previously unseen conditions. This setting is common in clinical practice but has received limited attention in existing research [8].

Addressing the challenges of zero-shot deployment requires methods that can generalize beyond the variations captured in training datasets. While state-of-the-art detectors such as YOLO maintain strong performance in general zero-shot detection tasks, their effectiveness can diminish in medical imaging scenarios where anatomical structures are complex, imaging conditions are highly variable, and acquisition artifacts are present [23]. In this context, VLMs have emerged as a promising complement, providing high-level semantic understanding that enhances the local feature representations of conventional detectors [15]. By integrating VLMs, detection systems can better interpret anatomical structures and distinguish polyps from complex backgrounds in scenarios not observed during training. This combination offers a practical strategy to improve generalization across unseen clinical centers, diverse imaging modalities, and acquisition conditions that are difficult to exhaustively annotate [1].

To evaluate detection under realistic, previously unseen endoscopic conditions, we simulate open-world scenarios by generating degraded variants of CVC-ClinicDB [4] and Kvasir-SEG [12] using Qwen-Image-Edit [28]. The synthetic images reflect clinically relevant variations, including dim illumination, mucus and fecal occlusions, and bubble interference. We propose a cascaded detector-verifier framework for robust polyp detection, where a YOLOv11

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detector first generates candidate bounding boxes that are subsequently refined by a VLM acting as a semantic verifier with adaptive, per-frame thresholding. This design improves recall while maintaining precision, mitigating clinically significant false negatives. To further enhance robustness under challenging imaging conditions, the VLM is fine-tuned using GRPO [24] to penalize missed detections more heavily than false positives, thereby emphasizing recall under challenging imaging conditions.

- Construction of a synthetic endoscopic dataset that simulates adverse conditions such as low illumination, bubbles, mucus, and fecal occlusion, providing a benchmark for evaluating polyp detection under realistic deployment scenarios.
- Development of a cascaded detector-verifier framework that adaptively adjusts detection thresholds to improve recall while maintaining precision for polyp detection in open-world settings.
- Introduction of a GRPO-based fine-tuning strategy to enhance detection reliability under challenging clinical conditions.

2. Related Works

Polyp Detection in Endoscopy. Polyp detection methods aim to enable real-time lesion localization by leveraging efficient, single-stage architectures that provide high throughput and low latency. Models such as YOLOv11 [13] exemplify this approach, employing lightweight feature extraction and multi-scale prediction mechanisms to balance detection accuracy with computational efficiency. The lightweight design and hardware-friendly deployment make these detectors suitable for high-frame-rate clinical environments such as live colonoscopy. However, despite their strong performance on standard datasets and general zero-shot tasks, conventional detectors can struggle in medical imaging scenarios where anatomical structures are complex and imaging conditions are highly variable [21, 25]. Domain shifts induced by motion blur, illumination changes, or occlusions frequently degrade detection reliability, potentially causing missed polyps or false alarms [4, 21, 25]. This limitation motivates the development of frameworks that combine fast, lightweight detectors with higher-level reasoning mechanisms, enabling adaptive confidence calibration and improved robustness under unpredictable real-world conditions [6, 27].

VLMs for Visual Reasoning. VLMs and multimodal large language models (MLLMs) have introduced a new paradigm for visual understanding by aligning visual features with linguistic representations through large-scale pre-training [22]. Models such as Qwen2.5-VL [26] exhibit strong semantic reasoning and promising zero-shot generalization. Nevertheless, these models often incur high computational cost and lack the spatial precision required for

pixel-level localization. They may also produce hallucinated outputs, generating semantically plausible but visually inconsistent predictions [11]. Such limitations indicate that pretraining alone is insufficient for reliable, safety-critical applications like polyp detection. Recent studies attempt to integrate structural priors from detectors with multimodal reasoning models. Frameworks such as LLaVA [16, 17] incorporate localized visual features into language backbones, improving visual grounding and diagnostic reliability. However, most existing approaches are evaluated on static, high-quality images and do not explicitly consider detection robustness under degraded endoscopic conditions [7]. This gap underscores the need for adaptive interaction between efficient detectors and VLMs to achieve reliable polyp detection in real-world clinical environments.

3. The Proposed Cascaded Detector-Verifier Framework

The proposed framework aims to improve polyp detection recall under previously unseen or degraded endoscopic conditions by combining a conventional detector with a VLM verifier and reinforcement learning-based optimization. Depicted in Fig. 1, an object detector initially generates candidate polyp bounding boxes, guided by adaptive thresholds set according to the VLM’s assessment of each frame. The VLM then verifies each candidate, refining detections and calibrating confidence to prioritize sensitivity. To further enhance robustness, the VLM is further optimized via GRPO, which emphasizes missed detections and calibrates confidence, enhancing reliability in low-contrast, noisy, or artifact-laden scenarios. Together, the cascaded architecture and GRPO fine-tuning form a unified system that balances recall and precision while aligning with clinically relevant objectives. The specifics of each component are elaborated next.

3.1. Cascaded Detector-Verifier with Semantic Thresholding and Verification

Semantic Thresholding for Candidate Generation. Each endoscopic frame is first evaluated by the VLM to assess overall imaging quality. The VLM predicts whether adverse conditions (e.g., dim illumination, motion blur, mucus, or bubble interference) are present, together with a confidence score. Based on this global assessment, the detection threshold for the YOLOv11 detector is adaptively selected. For frames classified as *Adverse*, a lower threshold $\theta_{c,low}$ is applied to favor recall under challenging conditions. For frames classified as *Clean*, a higher threshold $\theta_{c,high}$ is used to suppress false positives. YOLO then generates candidate bounding boxes according to the selected threshold. This adaptive, context-sensitive strategy balances sensitivity and specificity without introducing additional trainable parameters.

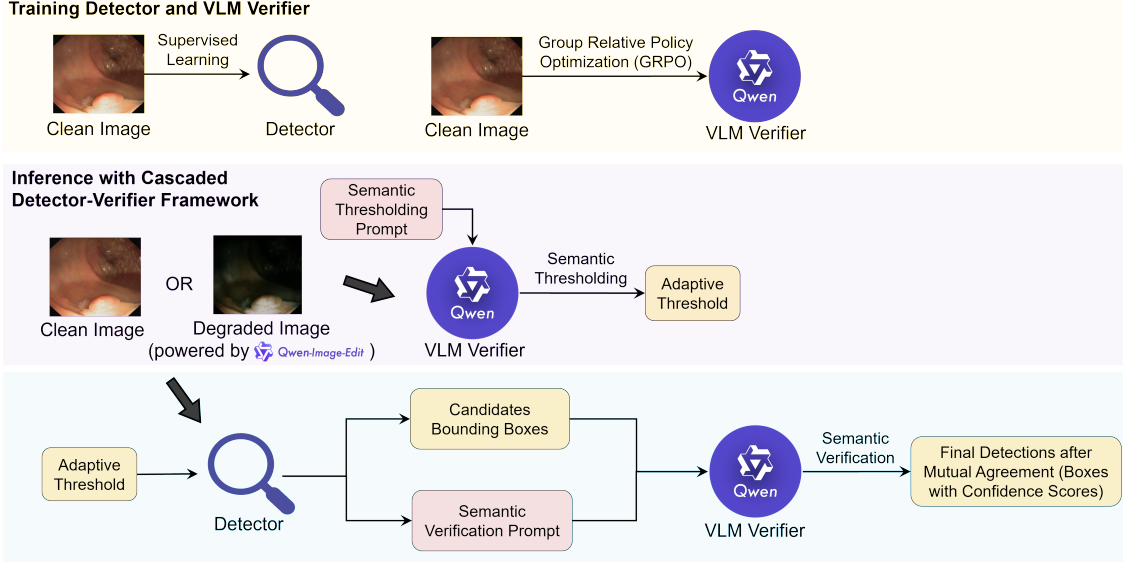


Figure 1. Overview of the proposed detector-verifier framework. Training: both the object detector and the VLM verifier are fine-tuned on clean images, with the detector using standard supervised learning and the verifier optimized via GRPO to penalize missed detections; inference: each image is first analyzed by the VLM to set an adaptive confidence threshold for the detector, producing candidate bounding boxes that reflect the VLM’s assessment of potential polyps. Each candidate is then verified by the VLM, and only those with mutual agreement and sufficient confidence are retained as final detections, ensuring sensitivity under challenging imaging conditions and improving robustness to previously unseen degradations.

Each endoscopic frame is first analyzed by the VLM, which predicts potential polyp locations along with confidence scores. Based on these predictions, the VLM sets the detection threshold for the YOLOv11 detector. In frames likely to contain polyps or exhibiting adverse imaging conditions, a lower threshold $\theta_{c,low}$ is applied to favor recall. In cleaner frames, a higher threshold $\theta_{c,high}$ is used to suppress false positives.

Semantic Verification and Mutual Agreement of Candidates. Each YOLO candidate is cropped and re-examined by the VLM using a fine-grained query. For k^{th} YOLO candidate in the i^{th} image. The VLM outputs binary decision d_{ik} along with a confidence c_{ik} to indicate mutual agreement or disagreement. For j^{th} polyp in the i^{th} image, detection success y_{ij} is evaluated as follows:

$$y_{ij} = \mathbb{1} \left[\max_{k \in \mathcal{K}_i} \left(\mathbb{1}[c_{ik} > \theta_{c,final}] \cdot d_{ik} \cdot \text{IoU}(B_{ik}, \hat{B}_{ij}) \right) \geq \theta_{iou} \right], \quad (1)$$

where $\mathcal{K}_i$ is the set of candidate boxes generated by YOLO for image I_i ; B_{ik} denotes the bounding box of the k^{th} YOLO candidate; $\hat{B}_{ij}$ is the ground-truth bounding box of the j^{th} polyp in the same image; $\text{IoU}(\cdot, \cdot)$ computes the Intersection-over-Union (IOU) between the candidate and the ground truth; $d_{ik} \in \{0, 1\}$ is the VLM verification decision, $\theta_{c,final}$ is the confidence threshold for confirming mutual agreement; θ_{iou} is the relaxed IoU threshold used to

count a polyp as successfully detected; and $\mathbb{1}[\cdot]$ is the indicator function, which outputs 1 if the condition is true and 0 otherwise. This formulation ensures that a ground-truth polyp is considered detected only if at least one YOLO candidate, confirmed by the VLM with sufficient confidence, sufficiently overlaps with the polyp according to the IoU threshold, formalizing mutual agreement between the detector and verifier while emphasizing sensitivity in clinically relevant scenarios.

3.2. Training of YOLO Detector and VLM Verifier

The YOLO detector and the VLM verifier are optimized with separate objectives and outputs. The YOLO detector is trained to generate candidate bounding boxes with high localization accuracy and objectness confidence, while the VLM verifier is trained using GRPO to generate and score its own candidate predictions. Each module focuses on its specialized task without constraining the other, and the two are only combined during inference, as defined in Sec. 3.1, to form the cascaded detector-verifier framework.

YOLO Detector Training. Following [13], the YOLO detector is fine-tuned to generate candidate bounding boxes with high localization accuracy and calibrated confidence scores by the VLM, balancing detection sensitivity and specificity for each frame.

VLM Verifier Optimization via GRPO. The VLM verifier is fine-tuned independently using GRPO to prioritize

missed detections while calibrating confidence. To provide a stable starting point, we first perform supervised fine-tuning (SFT) on paired image–annotation data, teaching the VLM basic grounding of polyps and ensuring reliable convergence [19]. Following SFT, GRPO evaluates multiple candidate outputs per image and guides the VLM toward relatively better predictions, improving robustness under low-contrast, occluded, or noisy conditions without requiring explicit exposure to degraded training data. The overall reward is defined as:

$$R_{\text{IoU}} = \frac{\sum_{i,j,k} \mathbb{1}[\text{IoU}(B_{ik}, \hat{B}_{ij}) \geq \theta_{\text{IoU}}] \cdot \text{IoU}(B_{ik}, \hat{B}_{ij})}{|\{(i, j, k) : \text{IoU}(B_{ik}, \hat{B}_{ij}) \geq \theta_{\text{IoU}}\}|}, \quad (2)$$

$$R_{\text{conf}} = \frac{1}{|\{(i, j, k)\}|} \sum_{i,j,k} \begin{cases} c_{ik}, & \text{if } \text{IoU}(B_{ik}, \hat{B}_{ij}) \geq \theta_{\text{IoU}}, \\ 1 - c_{ik}, & \text{otherwise,} \end{cases} \quad (3)$$

$$R_{\text{fn}} = \frac{1}{|\{(i, j)\}|} \sum_{i,j} \mathbb{1} \left[\max_{k \in \mathcal{K}_i} \text{IoU}(B_{ik}, \hat{B}_{ij}) < \theta_{\text{IoU}} \right], \quad (4)$$

$$R = \lambda_1 R_{\text{IoU}} + \lambda_2 R_{\text{conf}} + \lambda_3 R_{\text{format}} - \lambda_4 R_{\text{fn}}, \quad (5)$$

where $|\cdot|$ denotes the cardinality of a set; in R_{IoU} , the denominator counts valid matches of predictions and ground-truth objects, defined by the IoU constraints; in R_{conf} , the denominator counts all pairs of predictions and ground-truth objects; in R_{fn} , the denominator counts all ground-truth objects; λ_1 , λ_2 , λ_3 and λ_4 are hyperparameters that balance rewards' different influence during optimization; R_{IoU} promotes accurate localization by rewarding the VLM with predictions that largely overlap with ground-truth polyps; R_{conf} encourages calibrated confidence scores by rewarding high confidence for predictions that largely overlap with a ground-truth polyp, and low confidence otherwise; R_{format} is the format reward to ensure the outputs follow the required JSON schema [9]; R_{fn} counts the number of ground-truth polyps that are not sufficiently overlapped by any predicted bounding box (i.e., missed detections). This term penalizes false negatives by assigning a penalty whenever a polyp fails to match any prediction above the IoU threshold.

The GRPO-based reward explicitly emphasizes recall by guiding the VLM to better detect true polyps during training. The IoU component improves localization, while the confidence component calibrates scores, reinforcing reliable predictions and discouraging uncertain or spurious outputs. Although GRPO optimization during training and

candidate verification at inference are two distinct tasks with different inputs and prompts, prior studies [14, 20] have shown that reinforcement learning–based training can enhance a model’s downstream verification capabilities. As a result, the GRPO-trained VLM is more effective at confirming true YOLO candidates and rejecting false positives, improving overall detection reliability and supporting the clinical objective of minimizing missed lesions.

4. Experiment

Data. Two standard polyp detection benchmarks are used in this work: CVC-ClinicDB [4] and Kvasir-SEG [12]. The CVC-ClinicDB dataset contains 550 training images and 62 evaluation images, while Kvasir-SEG includes 900 training images and 100 evaluation images. All models are trained exclusively on the clean images from these datasets.

To evaluate robustness under realistic clinical conditions, we construct an additional synthetic testbed consisting of 500 degraded images using Qwen-Image-Edit [28]. Each degraded image is generated from clean endoscopy frames via prompt-based editing that simulates common colonoscopy artifacts, including dim illumination, motion blur, mucus or stool occlusion, and bubble interference. During generation, edit-time constraints are applied to preserve the original lesion annotations and spatial alignment. Note that the degraded dataset is never used during training, enabling a strict zero-shot evaluation on unseen adverse conditions.

Implementation Details. YOLOv11 detector series [13] are used and trained with an input resolution of 640×640 , a batch size of 16, for 100 epochs using SGD with momentum 0.937, weight decay 5×10^{-4} , an initial learning rate of 0.01 decayed via cosine annealing, and 3 warmup epochs.

The VLM verifier adopts Qwen3-VL-7B [3] and is first fine-tuned via SFT for 3 epochs with a per-device batch size of 4, gradient accumulation steps of 4 (effective batch size 16), a learning rate of 2×10^{-5} , and a maximum sequence length of 2,048 tokens. LoRA is applied with rank $r = 16$ and scaling factor $\alpha = 32$. For policy optimisation, GRPO is applied with a batch size of 2 prompts and a group size of 4, yielding 8 rollout sequences per step; gradient accumulation of 8 steps gives an effective prompt batch size of 16. The policy and value learning rates are 2×10^{-5} and 10^{-5} , respectively. The clipping parameter is 0.2, the discount factor is 0.99, and the GAE parameter is 0.95. The reward weights are $\lambda_1, \lambda_2, \lambda_3, \lambda_4 = 0.6, 0.3, 0.3, 2.0$. Note that for VLM SFT and GRPO fine-tuning, only the projection and text heads are updated while the backbone remains largely frozen, following the low-rank adaptation strategy in [10].

During inference, the detector operates with an initial confidence threshold of $\theta_{\text{c,high}} = 0.5$, which is lowered to $\theta_{\text{c,low}} = 0.2$ when the VLM indicates degraded image quality. The confidence and relaxed IoU thresholds are set

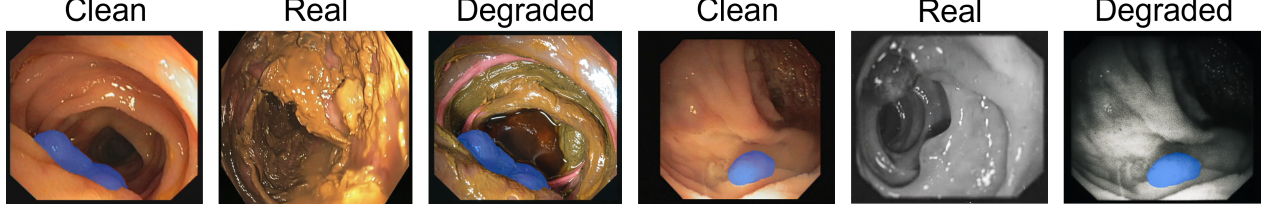


Figure 2. Examples of visualizations for synthetic degradations generated using Qwen-Image-Edit. The left three images comparatively show mucus and fecal occlusion, and the right three are for low illumination and bubble interference. For each set, the first column presents the clean source image with ground-truth annotation, the second column shows a real-world degraded image (annotations unavailable), and the third column shows the corresponding synthetically degraded image.

Table 1. Prompt templates used for semantic thresholding, computing detection results for GRPO fine-tuning, and semantic verification.

Task	Template
Semantic Threshold	Assess the overall quality of this endoscopic image and determine whether adverse imaging conditions are present (e.g. dim illumination, motion blur, mucus, or bubble interference). Return a global assessment (“Adverse” / “Clean”) and a confidence $\in [0, 1]$ (two decimals). Answer format: $\langle think \rangle \dots \langle /think \rangle \langle answer \rangle [\{ 'Assessment': 'Adverse/Clean', 'Confidence': c \}] \langle /answer \rangle$
Detection for GRPO	Detect all objects of class {polyp} in the image. Return a list of bounding boxes with integer coordinates $(x_1, y_1, x_2, y_2) \in [0, 1000]$ and a confidence $\in [0, 1]$ (two decimals). If no object exists, return “No Objects”. Answer format: $\langle think \rangle \dots \langle /think \rangle \langle answer \rangle [\{ 'Position': [x1,y1,x2,y2], 'Confidence': c \}, \dots] \langle /answer \rangle$
Semantic Verification	Examine the cropped region and decide if it contains a {polyp}. Return a binary decision (“Yes” / “No”) and a confidence $\in [0, 1]$ (two decimals). Answer format: $\langle think \rangle \dots \langle /think \rangle \langle answer \rangle [\{ 'Decision': 'Yes/No', 'Confidence': c \}] \langle /answer \rangle$

with $\theta_{c,final} = 0.7$ and $\theta_{iou} = 0.3$, respectively, for mutual agreement between the YOLO detector and the VLM verifier. The prompt templates used for semantic thresholding, computing detection results for GRPO fine-tuning, and semantic verification are summarized in Tab. 1. All experiments were conducted on an NVIDIA A800 GPU.

Evaluation Metrics. We evaluate detection performance using a binary criterion defined at the polyp level. Let $\hat{y}_{ij} \in \{0, 1\}$ denote the ground-truth label indicating whether the j^{th} polyp is present in the i^{th} image. Our framework predicts a detection indicator y_{ij} as defined in Eq. (1). Precision and Recall are computed in the conventional way by comparing the ground-truth $\hat{y}_{ij}$ with the prediction y_{ij} over all images and polyps.

To assess localisation quality beyond binary outcomes, we additionally report mean Intersection over Union (mIoU), defined as the average IoU over all valid matches:

$$\text{mIoU} = \frac{\sum_{i,j,k} \mathbb{1}[\text{IoU}(B_{ik}, \hat{B}_{ij}) \geq \theta_{iou}] \cdot \text{IoU}(B_{ik}, \hat{B}_{ij})}{|\{(i, j, k) : \text{IoU}(B_{ik}, \hat{B}_{ij}) \geq \theta_{iou}\}|}, \quad (6)$$

4.1. Synthetic Degradation Analysis

We quantitatively and qualitatively evaluate the perceptual fidelity of the generated synthetic degradations. In terms of metrics, fecal contamination achieves an Inception Score (IS) of 2.4 ± 0.7 and a Fréchet Inception Distance (FID) of 61.9, while low illumination with motion blur achieves an IS of 2.3 ± 0.1 and FID of 55.7. These results confirm the overall quality of the generation, indicating that both degradation types are perceptually faithful to real clinical examples. The low FID for illumination and motion blur reflects the simplicity of these transformations, while the high IS variance for fecal contamination indicates greater texture diversity, consistent with the semi-liquid, heterogeneous appearances observed in clinical samples.

Representative examples are provided in Fig. 2. The synthetic fecal contamination convincingly reproduces heterogeneous textures, color variations, and fluid-like structures, while the low illumination with motion blur accurately simulates reduced brightness and directional streaking. These results demonstrate that the generated degradations effectively capture the essential perceptual characteristics of real-world clinical endoscopy challenges, support-

ing their suitability for zero-shot evaluation.

4.2. Performance Comparison

Precision and Recall. Tab. 2 and Tab. 3 present the precision and recall comparisons for the YOLO detectors and our cascaded detector-verifier framework across the dirty (synthetically degraded) and clean subsets of CVC-ClinicDB and Kvasir-SEG. On the dirty data, the proposed framework consistently improves recall significantly across all detector variants. Averaged across detector variants, recall increases from 0.477 to 0.701 on CVC-ClinicDB and from 0.762 to 0.920 on Kvasir-SEG, demonstrating a substantial boost in sensitivity. These improvements highlight the zero-shot ability of our framework to detect challenging cases in degraded conditions, where YOLO alone tends to miss more detections. Also, as the detector size increases, the magnitude of recall improvement gradually decreases, suggesting that the proposed framework is more effective for smaller detectors.

On the clean data, the average recall improves more modestly, from 0.620 to 0.648 on CVC-ClinicDB and from 0.870 to 0.889 on Kvasir-SEG. Despite less pronounced, these improvements still show that the cascaded framework can enhance detection sensitivity even when the input images are of higher quality. This suggests that the framework’s performance is not limited to only challenging, degraded conditions but can also refine detection sensitivity across standard datasets, highlighting its versatility in medical applications where both degraded and standard conditions may be encountered.

While our framework improves recall, it results in a slight reduction in precision. On the dirty datasets, precision decreases from 0.752 to 0.741 on CVC-ClinicDB Dirty and from 0.878 to 0.872 on Kvasir-SEG Dirty. Similarly, on the clean datasets, precision drops from 0.803 to 0.796 on CVC-ClinicDB Clean and from 0.906 to 0.898 on Kvasir-SEG Clean. However, these slight reductions in precision represent a minor trade-off and is acceptable given the substantial gains in recall, especially in the medical context where prioritizing sensitivity is more critical.

Mean Intersection over Union. Tab. 4 compares the mIoU performance between the YOLO detectors and our cascaded detector-verifier framework. On the dirty subsets, our framework consistently improves mIoU for all detector variants, with the average increasing from 0.602 to 0.719 on CVC-ClinicDB and from 0.815 to 0.896 on Kvasir-SEG. On the clean subsets, mIoU improvements are smaller but still consistent, with average gains from 0.694 to 0.714 on CVC-ClinicDB and from 0.885 to 0.893 on Kvasir-SEG. These results demonstrate that our framework not only increases sensitivity but also enhances localization quality, particularly in challenging, degraded conditions.

Table 2. Precision and recall comparisons between YOLO detectors and our cascaded detector-verifier framework on the Dirty (synthetically degraded) subsets of CVC-ClinicDB and Kvasir-SEG. Note that our performance changes along with detector variants.

System	CVC-ClinicDB Dirty		Kvasir-SEG Dirty	
	Precision	Recall	Precision	Recall
YOLOv11n	0.781	0.512	0.870	0.789
Ours	0.745	0.748	0.846	0.936
YOLOv11s	0.836	0.498	0.894	0.801
Ours	0.812	0.736	0.887	0.945
YOLOv11m	0.701	0.489	0.904	0.738
Ours	0.708	0.726	0.912	0.932
YOLOv11l	0.752	0.534	0.885	0.712
Ours	0.769	0.754	0.872	0.889
YOLOv11x	0.690	0.355	0.839	0.770
Ours	0.672	0.543	0.847	0.902
YOLO Avg	0.752	0.477	0.878	0.762
Ours Avg	0.741	0.701	0.872	0.920

Table 3. Precision and recall comparisons between YOLO detectors and our cascaded detector-verifier framework on the Clean (original test) subsets of CVC-ClinicDB and Kvasir-SEG. Note that our performance changes along with detector variants.

System	CVC-ClinicDB Clean		Kvasir-SEG Clean	
	Precision	Recall	Precision	Recall
YOLOv11n	0.835	0.659	0.897	0.883
Ours	0.812	0.702	0.872	0.911
YOLOv11s	0.891	0.648	0.918	0.919
Ours	0.872	0.681	0.903	0.938
YOLOv11m	0.756	0.635	0.935	0.865
Ours	0.764	0.653	0.939	0.886
YOLOv11l	0.801	0.693	0.913	0.802
Ours	0.808	0.724	0.918	0.824
YOLOv11x	0.733	0.468	0.867	0.883
Ours	0.728	0.482	0.861	0.889
YOLO Avg	0.803	0.620	0.906	0.870
Ours Avg	0.796	0.648	0.898	0.889

4.3. Ablation Studies

We conduct ablation experiments to assess the contributions of individual system components on the Dirty sets of CVC-ClinicDB and Kvasir-SEG, with precision and recall results summarized in Tab. 5. Starting from the YOLO baseline, incorporating fixed thresholding (FT) 0.5 for confidence score improves both precision and recall, demonstrating the bene-

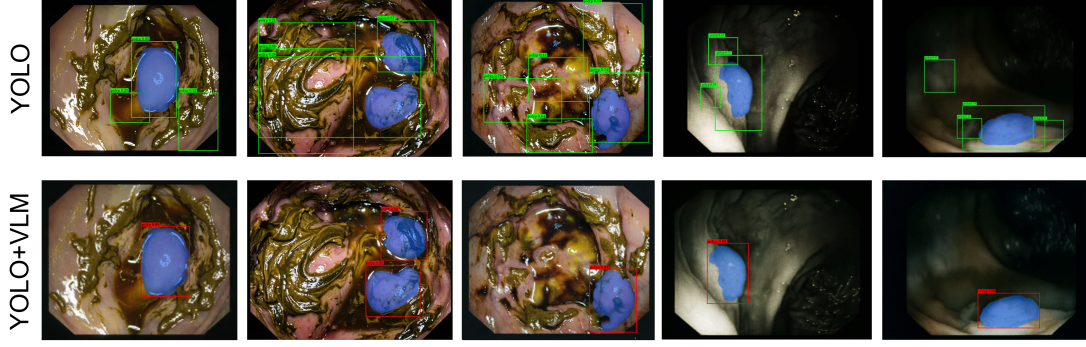


Figure 3. Zero-shot detection visualization on synthetically degraded frames. The left three columns correspond to samples of fecal occlusion, while the other columns correspond to samples of low illumination with motion blur. The first row shows YOLOv11s baseline detections, and the second row shows results from our cascaded detector-verifier framework. Green boxes denote YOLO detections, red boxes denote Qwen3-VL predictions, and blue regions indicate ground-truth masks.

Table 4. mIoU comparisons between YOLO detectors and our cascaded detector-verifier framework across Clean and Dirty subsets of CVC-ClinicDB and Kvasir-SEG.

	CVC Dirty	Kvasir Dirty	CVC Clean	Kvasir Clean
YOLOv11n	0.630	0.827	0.725	0.890
Ours	0.746	0.889	0.751	0.891
YOLOv11s	0.648	0.845	0.743	0.910
Ours	0.770	0.915	0.759	0.920
YOLOv11m	0.582	0.812	0.690	0.899
Ours	0.717	0.922	0.705	0.912
YOLOv11l	0.642	0.789	0.740	0.854
Ours	0.762	0.880	0.765	0.870
YOLOv11x	0.511	0.803	0.574	0.875
Ours	0.600	0.874	0.590	0.875
YOLO Avg	0.602	0.815	0.694	0.885
Ours Avg	0.719	0.896	0.714	0.893

fit of simple threshold adjustment. Adding semantic thresholding (ST) combined with semantic verification (SV) further enhances recall, highlighting the effectiveness of semantic reasoning by the VLM verifier. Finally, adding our proposed GRPO fine-tuning consistently yields the highest recall, with 75.4% on CVC-ClinicDB and 94.5% on Kvasir-SEG, while maintaining competitive precision. These results confirm that each component contributes to improved detection performance.

4.4. Detection Visualization

Fig. 3 qualitatively demonstrates the advantages of our cascaded detector-verifier framework over the YOLOv11s baseline on synthetically degraded frames. Our framework consistently reduces false negatives and suppresses spurious detections, while refining localization in challenging scenarios such as fecal occlusion, low illumination, and motion blur. For example, in heavily occluded frames, the VLM

Table 5. Precision and recall in different system settings on the Dirty sets on CVC-ClinicDB with YOLOv11l detector, and on Kvasir-SEG with YOLOv11s detector. FT: fixed thresholding; ST: semantic thresholding; SV: semantic verification.

Setting	CVC-ClinicDB		Kvasir-SEG	
	Precision	Recall	Precision	Recall
YOLO baseline	75.2	53.4	89.4	80.1
+ FT	77.4	67.0	90.0	88.0
+ ST + SV	77.2	72.0	89.2	92.0
+ ST + SV + GRPO (Ours)	76.9	75.4	88.7	94.5

verifier recovers lesions missed by YOLO, directly reducing clinically critical false negatives (column 2). In cases with false-positive candidates (column 1-5) or poor localization (column 2-5), the verifier confirms true polyps and rejects spurious boxes, yielding tighter and more reliable detections. These results qualitatively corroborate our performance improvements from integrating detector outputs with high-level semantic verification.

4.5. Computational Efficiency

We evaluate the computational efficiency of the proposed framework on an NVIDIA A800 GPU. During inference, the system requires approximately 16 GB of GPU memory and runs at 5 FPS. Although this throughput is lower than that of lightweight single-stage detectors, it remains applicable for real clinical colonoscopy workflows, where frames are typically sampled rather than processed continuously for analysis. Under this setting, a processing rate of 5 FPS is sufficient to support downstream detection and verification. Nevertheless, the current computational cost reflects a limitation of the cascaded detector-verifier design. Improving inference efficiency while maintaining detection robustness under challenging degradations will be an important direction for our future work.

5. Conclusion

This paper presents a cascaded detector and verifier framework for zero-shot polyp detection in open-world endoscopy, combining a YOLOv11 detector with a vision–language model verifier that applies semantic thresholding and verification. The VLM verifier is optimized via GRPO to penalize missed detections and align with clinically relevant objectives. We also introduce synthetically generated degraded datasets that simulate illumination changes, motion blur, and fecal contamination, providing a benchmark for evaluating zero-shot robustness under challenging imaging conditions. Extensive experiments on CVC-ClinicDB, Kvasir-SEG, and these degraded datasets demonstrate significant recall improvements while maintaining precision comparable to the YOLO baseline, highlighting enhanced localization and detection robustness across detector variants. Our future work will focus on improving inference efficiency and extending the framework to real-world endoscopic video streams.

References

- [1] Sharib Ali, Noha Ghatwary, Debesh Jha, Ece Isik-Polat, Gorkem Polat, Chen Yang, Wuyang Li, Adrian Galdran, Miguel-Ángel González Ballester, Vajira Thambawita, et al. Assessing generalisability of deep learning-based polyp detection and segmentation methods through a computer vision challenge. *Scientific Reports*, 14(1):2032, 2024. 1
- [2] Salih Aliyi, Kokeb Dese, and Hakkins Raj. Detection of gastrointestinal tract disorders using deep learning methods from colonoscopy images and videos. *Scientific African*, 20: e01628, 2023. 1
- [3] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhifang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhao-hai Li, Mingsheng Li, Mei Li, Kaixin Li, Zicheng Lin, Junyang Lin, Xuejing Liu, Jiawei Liu, Chenglong Liu, Yang Liu, Dayiheng Liu, Shixuan Liu, Dunjie Lu, Ruilin Luo, Chenxu Lv, Rui Men, Lingchen Meng, Xuancheng Ren, Xingzhang Ren, Sibao Song, Yuchong Sun, Jun Tang, Jianhong Tu, Jianqiang Wan, Peng Wang, Pengfei Wang, Qiuyue Wang, Yuxuan Wang, Tianbao Xie, Yiheng Xu, Haiyang Xu, Jin Xu, Zhibo Yang, Mingkun Yang, Jianxin Yang, An Yang, Bowen Yu, Fei Zhang, Hang Zhang, Xi Zhang, Bo Zheng, Humen Zhong, Jingren Zhou, Fan Zhou, Jing Zhou, Yuanzhi Zhu, and Ke Zhu. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025. 4
- [4] Jorge Bernal, F Javier Sánchez, Gloria Fernández-Esparrach, Debora Gil, Cristina Rodríguez, and Fernando Vilariño. Wm-dova maps for accurate polyp highlighting in colonoscopy: Validation vs. saliency maps from physicians. *Computerized medical imaging and graphics*, 43:99–111, 2015. 1, 2, 4
- [5] Pedro Carrinho and Gabriel Falcao. Highly accurate and fast yolov4-based polyp detection. *Expert Systems with Applications*, 232:120834, 2023. 1
- [6] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018. 2
- [7] Christoph Feichtenhofer, Haoqi Fan, Jitendra Malik, and Kaiming He. Slowfast networks for video recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 6202–6211, 2019. 2
- [8] Hao Guan and Mingxia Liu. Domain adaptation for medical image analysis: A survey. *IEEE Transactions on Biomedical Engineering*, 69(3):1173–1185, 2022. 1
- [9] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025. 4
- [10] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. 4
- [11] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. A survey on hallucination in large language models. *arXiv preprint arXiv:2311.05232*, 2023. 2
- [12] Debesh Jha, Pia H Smedsrud, Michael A Riegler, Pål Halvorsen, Thomas de Lange, Dag Johansen, and Håvard D Johansen. Kvasir-seg: A segmented polyp dataset. In *MultiMedia Modeling: 26th International Conference, MMM 2020, Daejeon, South Korea, January 5–8, 2020, Proceedings, Part II 26*, pages 451–462, 2020. 1, 4
- [13] Glenn Jocher and Jing Qiu. Ultralytics yolo11. <https://github.com/ultralytics/ultralytics>, 2024. 2, 3, 4
- [14] Yuxiang Lai, Jike Zhong, Ming Li, Shitian Zhao, Yuheng Li, Konstantinos Psounis, and Xiaofeng Yang. Med-r1: Reinforcement learning for generalizable medical reasoning in vision-language models. *IEEE Transactions on Medical Imaging*, 2026. 4
- [15] Xiang Li, Like Li, Yuchen Jiang, Hao Wang, Xinyu Qiao, Ting Feng, Hao Luo, and Yong Zhao. Vision-language models in medical image analysis: From simple fusion to general large models. *Information Fusion*, 118:102995, 2025. 1
- [16] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023. 2
- [17] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. *arXiv preprint arXiv:2310.03744*, 2024. 2
- [18] Yassine Oukdach, Anass Garbaz, Zakaria Kerkaou, Mohamed El Ansari, Lahcen Koutti, Ahmed Fouad El Ouafdi, and Mouna Salihoun. Uvit-seg: an efficient vit and u-net-based framework for accurate colorectal polyp segmentation in colonoscopy and wce images. *Journal of Imaging Informatics in Medicine*, 37(5):2354–2374, 2024. 1

- [19] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, pages 27730–27744. Curran Associates, Inc., 2022. [4](#)
- [20] Jiazhen Pan, Che Liu, Junde Wu, Fenglin Liu, Jiayuan Zhu, Hongwei Bran Li, Chen Chen, Cheng Ouyang, and Daniel Rueckert. Medvlm-r1: Incentivizing medical reasoning capability of vision-language models (vlms) via reinforcement learning. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 337–347. Springer, 2025. [4](#)
- [21] Konstantin Pogorelov, Kristin Ranheim Randel, Carsten Grieder, Concetto Spampinato, Thomas de Lange, Dag Johansen, and Michael Riegler. Kvasir: A multi-class image dataset for computer aided gastrointestinal disease detection. In *Proceedings of the 8th ACM on Multimedia Systems Conference*, pages 164–169, 2017. [2](#)
- [22] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763. PMLR, 2021. [2](#)
- [23] Mohammed Gamal Ragab, Said Jadid Abdulkadir, Amgad Muneer, Alawi Alqushaibi, Ebrahim Hamid Sumiea, Rizwan Qureshi, Safwan Mahmood Al-Selwi, and Hitham Alhusian. A comprehensive systematic review of yolo for medical object detection (2018 to 2023). *IEEE Access*, 12:57815–57836, 2024. [1](#)
- [24] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. [2](#)
- [25] Nima Tajbakhsh, Suryakanth R Gurudu, and Jianming Liang. Automated polyp detection in colonoscopy: Lessons learned from a piggyback approach and 15 years of data. *Computerized Medical Imaging and Graphics*, 46:235–243, 2015. [2](#)
- [26] Qwen Team. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025. [2](#)
- [27] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017. [2](#)
- [28] Chenfei Wu, Jiahao Li, Jingren Zhou, Junyang Lin, Kaiyuan Gao, Kun Yan, Sheng ming Yin, Shuai Bai, Xiao Xu, Yilei Chen, Yuxiang Chen, Zecheng Tang, Zekai Zhang, Zhengyi Wang, An Yang, Bowen Yu, Chen Cheng, Dayiheng Liu, Deqing Li, Hang Zhang, Hao Meng, Hu Wei, Jingyuan Ni, Kai Chen, Kuan Cao, Liang Peng, Lin Qu, Minggang Wu, Peng Wang, Shuting Yu, Tingkun Wen, Wensen Feng, Xiaoxiao Xu, Yi Wang, Yichang Zhang, Yongqiang Zhu, Yujia Wu, Yuxuan Cai, and Zenan Liu. Qwen-image technical report, 2025. [1](#), [4](#)
- [29] Huisi Wu, Zebin Zhao, Jiafu Zhong, Wei Wang, Zhenkun Wen, and Jing Qin. Polypseg+: A lightweight context-aware network for real-time polyp segmentation. *IEEE Transactions on Cybernetics*, 53(4):2610–2621, 2022. [1](#)